Assembly Maps in Algebraic K-theory

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✿ Introduction

Given a ring R (associative, unital), one can assign a sequence $\{K_n(R)\}_{n=1}^{\infty}$ of Abelian groups, called the algebraic K-theory of R:

$$K_n(R) := \pi_n(K(R)).$$

where $K(R)$ is the K-theory spectrum of R, built in a way to reflect structures of the category of finitely generated projective left R-modules, after Quillen (1973), Waldhausen (1985).

✿ The Original Definition of $K_0(R)$ (Grothendieck 1950's)

Let $\mathrm{Proj}(R)$ be the set of equivalence classes of finitely generated projective left R-modules, where $M \sim N$ iff $M \cong N$.

$(\mathrm{Proj}(R), \oplus)$ is a commutative monoid. $K_0(R)$ is defined to be the group completion of $(\mathrm{Proj}(R), \oplus)$. (Grothendieck group)

$\boxed{\text{e.g.}}$ $K_0(\text{field}) \cong \mathbb{Z}$.

$K_0(\text{PID}) \cong \mathbb{Z}$ (follows from the Structure Theorem) ▨

❀ Functoriality of K_0

If $\varphi: R \to S$ is a ring homomorphism, then φ induces a map $\tilde{\varphi}: \mathrm{Proj}(R) \to \mathrm{Proj}(S)$ by

$$[P] \mapsto [S \otimes_\varphi P].$$

Here, S is consider as a right R-module by $s \cdot r := s\varphi(r)$.

One can check $S \otimes_\varphi P$ is indeed finitely generated projective: if $P \oplus Q \cong R^n$,

$$\begin{aligned}(S \otimes_\varphi P) \oplus (S \otimes_\varphi Q) &\cong S \otimes_\varphi (P \oplus Q) \\ &\cong S \otimes_\varphi R^n \\ &\cong S^n.\end{aligned}$$

Functoriality of Grothendieck group gives $\varphi_*: K_0(R) \to K_0(S)$.

❀ The Original Definition of $K_1(R)$ (JHC Whitehead 1939, 1941, 1950)

Define $GL(n,R) \hookrightarrow GL(n+1,R)$ by $A \mapsto \begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix}$.

We can then define $GL(R) := \bigcup_{n=1}^{\infty} GL(n,R)$, and

$$K_1(R) := GL(R)^{ab} \qquad \text{(Abelianisation)}$$

$\boxed{\text{e.g.}}$ If R is an Euclidean domain, then $K_1(R) \cong R^\times$. ▨

❀ Goal

Understand $K_n(R[G])$.

For any group G, the group ring $R[G]$ is defined to be

$$R[G] := \left\{ \text{formal sum } \sum_{g\in G} r_g g \;\middle|\; r_g \in R \right\}.$$

with multiplication $(r_g g)\cdot(r_{g'} g') = (r_g r_{g'})\, gg'$.

$\boxed{\text{e.g.}}$ $R[\mathbb{Z}] \cong R[t^{\pm 1}]$,

$R[\mathbb{Z}/p\mathbb{Z}] \cong R[t]/\langle t^p - 1\rangle$.

$\boxed{\mathbb{Z}}$

There is the Loday assemby map

$$(\alpha_L)_n : H_n(BG; K(R)) \to K_n(R[G])$$

which approximates the K-theory of $R[G]$ by a homology theory with coefficients in $K(R)$.

$(\alpha_L)_n$, however, is far from isomorphism.

So we need to consider subgroups of G.

✿ Family of Subgroups

If G is a group, by a family of subgroups of G we mean a set $\mathcal{F}$ of subgroups of G such that:

(i) if $H \in \mathcal{F}$, $K \leq H$, then $K \in \mathcal{F}$

(ii) if $H \in \mathcal{F}$, then $gHg^{-1} \in \mathcal{F}$ for all $g \in G$.

$\boxed{\text{e.g.}}$ $\mathcal{F} = ALL$, all subgroups

$\mathcal{F} = VC$, virtually cyclic subgroups

$\mathcal{F} = FIN$, finite subgroups

$\mathcal{F} = \{1\}$, trivial subgroup

✿ Orbit Category

$Or(G,\mathcal{F})$ is the category of

objects : G-sets G/H , where $H \in \mathcal{F}$

morphisms: G-equivariant maps $G/H_1 \to G/K_1$

We write $Or(G) := Or(G, ALL)$.

❀ Action Groupoid

We define a functor $G\int - : \mathrm{Or}(G) \longrightarrow \mathrm{GROUPOIDS}$ by

$$G/H \mapsto G\int G/H,$$

where $\mathrm{Obj}(G\int G/H) := G/H$ as a set, and

$$g_1H \xrightarrow{g\in G} g_2H, \text{ such that } (gg_1)H = g_2H.$$

So $G\int G/H$ is a groupoid, which we call the action groupoid of G/H.

❀ The Davis-Lück K-theory

Fix a ring R, Davis-Lück (1998) constructed a functor

$$K^{DL}: \text{GROUPOIDS} \to \Omega\text{-SPECTRA}$$

such that if $\mathcal{G}$ is a connected groupoid, and $\mathcal{G}_x \subseteq \mathcal{G}$ is a full sub-groupoid with only one object x, then

$$K^{DL}(\mathcal{G}) \simeq K^{DL}(\mathcal{G}_x).$$

In particular,

$$K^{DL}(\mathcal{G}_x) = K(R[\mathrm{Aut}_{\mathcal{G}}(x)]).$$

Now, if $\mathcal{G} = G \wr G/H$ is the action groupoid, and $x = H \in G/H$, then

$$
\begin{aligned}
K^{DL}(G \wr G/H) &= K^{DL}(\mathcal{G}) \\
&\simeq K^{DL}(\mathcal{G}_x) \\
&= K(R[\mathrm{Aut}_{G \wr G/H}(H)]) \\
&= K(R[\{g \in G \mid gH = H\}]) \\
&= K(R[H]).
\end{aligned}
$$

Therefore $\pi_n(K^{DL}(G \wr G/H)) \cong K_n(R[H])$.

We promote $K^{DL}(G\int -) : G\text{-SETS} \to \Omega\text{-SPECTRA}$ to a new functor

$$E_{\%} : G\text{-SETS} \longrightarrow \Omega\text{-SPECTRA}$$

by

$$X \longmapsto \operatorname{map}_G(-, X) \wedge_{Or(G)} K^{DL}(G\int -)$$

$$:= \left(\coprod_{G/H \in Or(G)} \operatorname{map}_G(G/H, X) \times K^{DL}(G \int G/H) \right) \Big/ \sim$$

where $(\phi^*(x), y) \sim (x, \phi_*(y))$ for

$\phi \in \operatorname{Mor}_{Or(G)}(G/H_1, G/H_2)$, and

$x \in \operatorname{map}_G(G/H_2, X)$, $y \in K^{DL}(G \int G/H_1)$.

In particular, when $X = \text{point}$, then

$$E_{\%}(\text{point}) := \left(\coprod_{G/H \in Or(G)} \text{map}_G(G/H, \text{point}) \times K^{DL}(G \int G/H) \right) / \sim$$

$$\cong \left(\coprod_{G/H \in Or(G)} K^{DL}(G \int G/H) \right) / \left(G/H \sim \phi(G/H) \right)$$

$$\cong K^{DL}(G \int G/G)$$

$$\cong K(R[G]).$$

We will choose X to be another special space.

❀ Classifying Space for a Family

Given $(G, \mathcal{F})$, there exists a G-CW-complex $E_{\mathcal{F}}G$, unique up to G-homotopy equivalence, characterised by

$$(E_{\mathcal{F}}G)^H \text{ is } \begin{cases} \text{contractible} & \text{if } H \in \mathcal{F}, \\ \emptyset & \text{if else}. \end{cases}$$

$\boxed{\text{e.g.}}$ $E_{ALL}G = \text{point}.$

$E_{\{1\}}G = EG.$

❀ The Assembly Map

The assembly map is a map of Ω-spectra

$$\alpha_{\mathcal{F}} : E_{\%}(E_{\mathcal{F}}G) \to E_{\%}(\text{point}) \cong K(R[G]).$$

induced by $E_{\mathcal{F}}G \to \text{point}$. Taking homotopy groups, we have

$$(\alpha_{\mathcal{F}})_n : H_n^G(E_{\mathcal{F}}G\,;K(R)) \longrightarrow K_n(R[G]).$$

The source $H_n^G(E_{\mathcal{F}}G; K^{DL}(R))$ is a G-homology theory, and can be computed by an equivariant version of the Atiyah-Hirzebruch spectral sequence:

$$E^2_{p,q} := H_p(C^c_*(E_{\mathcal{F}}G) \otimes_{\mathbb{Z}Or(G)} \pi_q(K^{DL}(G/-))) \quad \text{(Bredon Homology)}$$

$$\Rightarrow H_n^G(E_{\mathcal{F}}G; K(R))$$

However, the target $K_n(R[G])$ is difficult to compute.

The case $\mathcal{F} = \{1\}$ yields the Loday assembly map

$$(\alpha_L)_n : H_n(BG; K(R)) \to K_n(R[G]),$$

where the source is an ordinary homology theory with coefficients in the K-theory spectrum, and can be computed by the Atiyah-Hirzebruch spectral sequence

$$E^2_{s,t} := H_s(BG; K_t(R)) \Rightarrow H_{s+t}(BG; K(R)).$$

✿ The Farrell-Jones Conjecture (FJC)

$\boxed{\text{Conjecture}}$ (FJC) If $\mathcal{F} = VC$, then $(\alpha_{\mathcal{F}})_n$ is an isomorphism for all $n \geq 0$. ▨

Note that $(\alpha_{\mathcal{F}})_0 : K_0(R) \to K_0(R[G])$ is the group homomorphism induced by the ring inclusion $R \hookrightarrow R[G]$.

What We Know Already

FJC is true for

1) Hyperbolic groups

(Bartels-Lück-Reich, 2008)

2) CAT(0)-groups

(Bartels-Lück, 2009)

3) Solvable groups

(Wegner, 2013)

What We Do Not Know Yet

if FJC is true for

1) Linear groups

2) Infinite product of groups satisfying FJC

3) $Out(F_n)$

❀ An Application of FJC — the Kaplansky's Conjecture

Consider the algebra $\mathcal{M}_n := M(n,R)$. Say R is stably finite if for every $n \geq 1$, an $\mathcal{M}_n$-module N is trivial whenever $\mathcal{M}_n \cong \mathcal{M}_n \oplus N$.

$\boxed{\text{e.g.}}$ If R is a field with $\mathrm{char}(R)=0$, then $R[G]$ is stably finite for all groups G. (Kaplansky) ▨

$\boxed{\text{Lemma}}$ Let R be a ring with $\mathrm{Idem}(R)=\{0,1\}$, and G is torsion-free. If the map $i_*: K_0(R) \to K_0(R[G])$ is rationally bijective, and $R[G]$ is stably finite, then $\mathrm{Idem}(R[G])=\{0,1\}$. ▨

$\boxed{\text{Proof}}$ Define the argumentation map $\varepsilon : R[G] \to R$ by

$$\sum_{g \in G} r_g g \mapsto \sum_{g \in G} r_g .$$

If $p \in \mathrm{Idem}(R[G])$, then $\varepsilon(p) \in \mathrm{Idem}(R)$. So either $\varepsilon(p) = 0$ $\varepsilon(p) = 1$.

① If $\varepsilon(p) = 0$, the ideal $\langle p \rangle \trianglelefteq R[G]$, treated as a finitely generated projective $R[G]$-module:

$$R[G] \cong \langle p \rangle \oplus \{x \in R[G] \mid p \cdot x = 0\}.$$

The element $[\langle p \rangle] \otimes 1 \in K_0(R[G]) \otimes \mathbb{Q}$ has a pre-image

$$\sum_{i=1}^{n} [M_i] \otimes \frac{a_i}{b_i} \in K_0(R) \otimes \mathbb{Q}.$$

Write $b := \prod_{i=1}^{n} b_i$, $b_i' := \prod_{j \neq i} b_j$. Then in $K_0(R) \otimes \mathbb{Q}$,

$$b \cdot \sum_{i=1}^{n} [M_i] \otimes \frac{a_i}{b_i} = [\underbrace{\bigoplus_{i=1}^{n} M_i^{\oplus a_i b_i'}}_{M}] \otimes 1.$$

$$\Rightarrow (i_* \otimes \mathrm{id}_{\mathbb{Q}})([M] \otimes 1) = [\langle p \rangle^{\oplus b}] \otimes 1 \quad \in K_0(R[G]) \otimes \mathbb{Q}.$$

$$\Rightarrow i_*([M]) - [\langle p \rangle^{\oplus b}] \in \mathrm{Tors}(K_0(R[G])).$$

So there is $m>0$ such that $m\cdot\left(i_*([M]) - [\langle P\rangle^{\oplus b}]\right) = 0$, i.e.,

$$i_*([M^{\oplus m}]) = [\langle P\rangle^{\oplus bm}].$$

The relation used to construct the Grothendieck group then gives

$$\tilde{\iota}(M^{\oplus m}) \oplus R[G]^{\oplus n} \cong \langle P\rangle^{\oplus bm} \oplus R[G]^{\oplus n}$$

for some $n \geq 0$. One then look at the images of both sides under $\tilde{\iota}\circ\tilde{\varepsilon}$.

$$\boxed{\tilde{i}(M^{\oplus m}) \oplus R[G]^{\oplus n} \cong \langle p \rangle^{\oplus bm} \oplus R[G]^{\oplus n}}$$

Left Hand Side

$$\begin{aligned}
\tilde{i}\circ\tilde{\varepsilon}\left(\tilde{i}(M^{\oplus m}) \oplus R[G]^{\oplus n}\right) &\cong \tilde{i}\circ\tilde{\varepsilon}(\tilde{i}(M^{\oplus m})) \oplus \tilde{i}\circ\tilde{\varepsilon}(R[G]^{\oplus n}) \\
&\cong \tilde{i}(M^{\oplus m}) \oplus \tilde{i}\circ\tilde{\varepsilon}(R[G]^{\oplus n}) \quad (\tilde{\varepsilon}\circ\tilde{i} = id) \\
&\cong \tilde{i}(M^{\oplus m}) \oplus R[G] \otimes_{i\circ\varepsilon} R[G]^{\oplus n} \\
&\cong \tilde{i}(M^{\oplus m}) \oplus R[G]^{\oplus n}
\end{aligned}$$

Right Hand Side

$$\begin{aligned}
\tilde{i}\circ\tilde{\varepsilon}\left(\langle p \rangle^{\oplus bm} \oplus R[G]^{\oplus n}\right) &\cong R[G] \otimes_{i\circ\varepsilon} \langle p \rangle^{\oplus bm} \oplus R[G]^{\oplus n} \\
&\cong 0 \oplus R[G]^{\oplus n} \quad (\varepsilon(p) = 0) \\
&\cong R[G]^{\oplus n}
\end{aligned}$$

Conclude : $\langle p\rangle^{\oplus bm} \oplus R[G]^{\oplus n} \cong R[G]^{\oplus n}$.

Because $R[G]$ is stably finite by assumption, we must have $\langle p\rangle^{\oplus bm} = 0$. Consequently, $p = 0$.

We showed $p = 0$ whenever $\varepsilon(p) = 0$.

② If $\varepsilon(P)=1$, then notice that $1-P \in \mathrm{Idem}(R[G])$, and

$$
\begin{aligned}
0 &= 1-1 \\
&= \varepsilon(1)-\varepsilon(P) \\
&= \varepsilon(1-P).
\end{aligned}
$$

By ①, we must have $1-P=0$, hence $P=1$.

This completes the proof. $\blacksquare$

Conjecture (Kaplansky)

If R is an integral domain, and G torsion-free, then $\mathrm{Idem}(R[G]) = \{0,1\}$.

Thm Assume FJC is true, then KC is true when R is a field with $\mathrm{char}(R) = 0$, and G torsion-free.

Proof FJC $\Rightarrow$ $i_* = (\alpha_{\mathcal{F}})_0$ is rationally bijective.

R is a field with $\mathrm{char}(R) = 0$, so $R[G]$ is stably finite, and $\mathrm{Idem}(R) = \{0,1\}$. The claim then follows from the Lemma.

❀ Proposed Research Direction

① When $\mathcal{Y}=\{1\}$, $\alpha_{\mathcal{Y}}$ is the Loday assembly map. FJC predicts $(\alpha_{\{1\}})_n$ is an isomorphism for all $n \geq 0$ when R is regular and G is torsion-free.

Ullmann-Wu proved when $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and R be any field with $\mathrm{char}(R) > 2$, then $(\alpha_{\{1\}})_2$ is not injective. (2016)

<u>Question:</u> If $R = \mathbb{Z}$, is there a group G with non-trivial torsion, such that $(\alpha_{\{1\}})_n$ is not injective for some n?

② Hesselholt – Madsen used pointed monoid algebras to study K-theory of truncated polynomial algebras.

Question: Can one construct a Loday-like assembly map, and formulate an isomorphism conjecture for K-theory of pointed monoid algebras?

③ Can we prove FJC for new classes of groups?